

Appendix: Frontlight mod for Watchy v2 / USB-C clone

Supplement to `watchy-v3-frontlight-mod.md`. All chapters of the main document apply except 2 (circuit), 4 (connection points), 5.1 (BOM) and 6 (firmware) — those are replaced here.

Sources: `Watchy.sch` / `Watchy.kicad_pcb` rev 2.0 (official SQFMI layout), cross-checked against `WatchySchematic.pdf`.

A1. Why the v2 needs a different circuit

On the v3, TP2 (GPIO44) provides a free, switchable GPIO on a test pad. The v2 has no equivalent:

GPIO	Status in the layout
32	unconnected, no trace
2	unconnected, no trace
33	taken — net <code>32KHZ</code> (CLKO of the PCF8563, pin 8)
15	tied to GND
0	brought out to TP1, but a strapping pin

GPIO16/17 are internally wired to the embedded flash on the ESP32-PICO-D4 and are ruled out entirely.

TP1 (GPIO0) would be the only candidate. A pulldown there would force the watch into the bootloader at power-up; the only way around that is a P-channel high-side switch, and GPIO0 also hangs off the CP2102 auto-reset circuit. Most clones omit TP1 anyway.

Consequence: I²C LED driver instead of a MOSFET on a GPIO.

A2. Connection points

Four pads, 1 mm diameter, round, on the component side, arranged as a 2×2 field at 2.54 mm pitch. Everything the circuit needs sits in that one square.

TP	Net	Position in the original layout (mm)
TP2	SCL	91.38 / 105.82
TP3	SDA	91.38 / 108.36
TP4	+3V3	93.92 / 105.82
TP5	GND	93.92 / 108.36

Arrangement seen from the component side:

SCL	3V3
SDA	GND

Not part of the block and usually omitted on clones: TP1 (GPIO0) at 86.90 / 91.61 and TP6 (GND) at 82.85 / 92.16.

A2.1 Identifying the pads

Battery removed, powered over USB. Black probe on the shell of the USB connector.

Measurement	Pad
0.00 V	GND
3.23-3.37 V, steady	+3V3
approx. 4.7 kΩ to the 3V3 pad	SCL or SDA

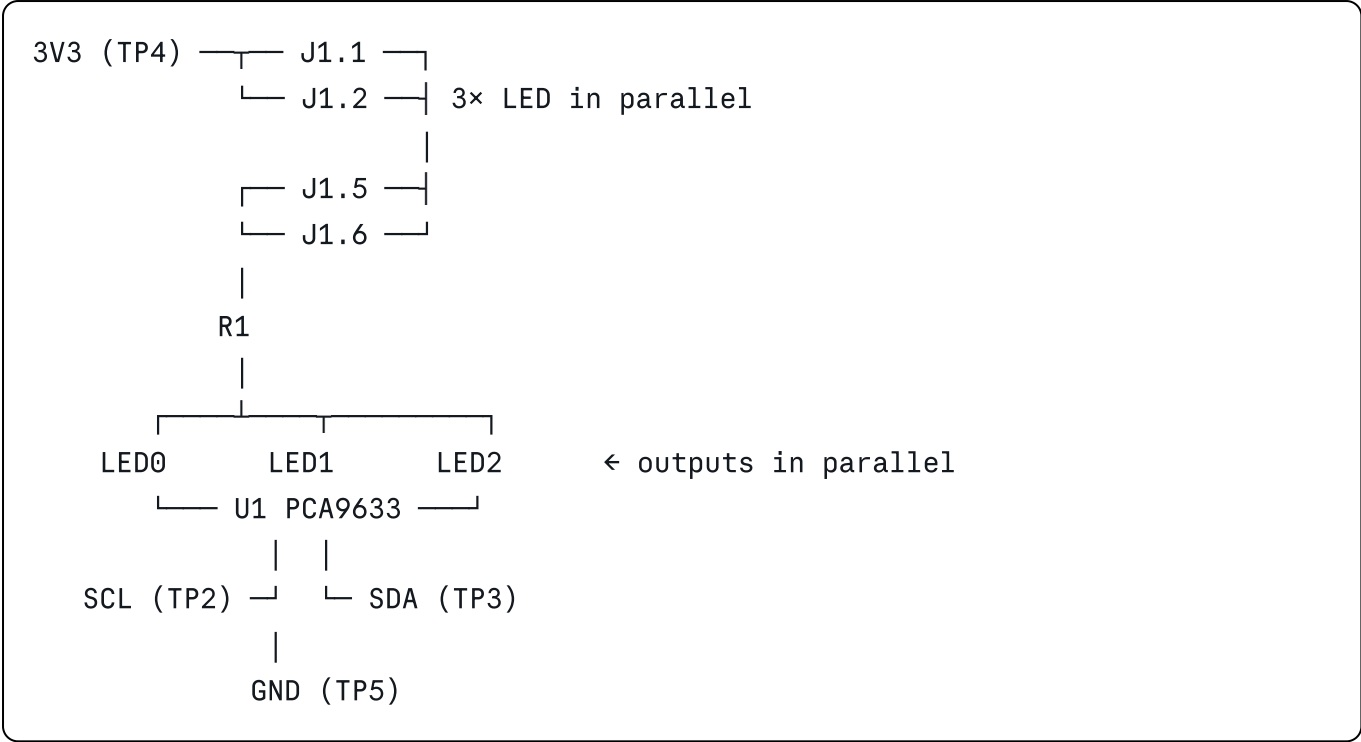
The pull-ups are R2 and R3, 4.7 kΩ each, to +3V3.

You do not need to know the orientation of the block. Once GND and 3V3 are identified, the remaining two are necessarily SCL and SDA. Swapping them does no harm — the I²C scanner simply will not find the driver, you swap the two wires, done. Unlike confusing 3V3 with battery voltage, this has no consequences.

A3. Circuit

Replaces sections 2.3 to 2.5 of the main document.

A3.1 Topology



A3.2 Netlist

Net	Connections
3V3	TP4 · J1 pin 1 · J1 pin 2 · U1 VDD · C1-A
LED_K	J1 pin 5 · J1 pin 6 · R1-A
SW	R1-B · U1 LED0 · U1 LED1 · U1 LED2
SCL	TP2 · U1 SCL
SDA	TP3 · U1 SDA
GND	TP5 · U1 VSS · C1-B

Four wires to the mainboard instead of three.

A3.3 Why three outputs in parallel

The PCA9633 channels are current sinks rated **25 mA each**. The frontlight draws up to 45 mA, and the three LED dies share a common cathode net inside the module — driving them individually is not possible.

Three channels in parallel work out to 15 mA per channel at 45 mA total. Even with uneven sharing between the sinks, every channel stays well inside its rating. Channel LED3 is left unused.

A3.4 I²C address

Device	Address
BMA423	0x18 or 0x19
PCF8563	0x51
PCA9633	0x62

No collision. The v2 bus is comfortably loaded with two devices and nowhere near capacitance limits.

A3.5 R1

Unchanged from sections 2.5 and 2.6 of the main document: **22 Ω as a starting value**, approach from above, never below 10 Ω . The voltage headroom is the same, because the v2 also runs a 3.3 V LDO (ME6211C33M5G-N, 500 mA).

The PCA9633 adds a small extra voltage drop — all the more reason to measure the current on the finished assembly rather than calculating it.

A4. Bill of materials

Replaces section 5.1.

Ref	Part	Specification	Qty
—	Display	GDEY0154D67-FL04	1
U1	LED driver	PCA9633, I ² C, 4× 25 mA	2
C1	Capacitor	100 nF, 0402	2
R1	Resistor	assortment 10/15/22/27/39/68 Ω , 0603	5 each
J1	FPC connector	6-pin, 0.5 mm, ZIF, top contact, 0.3 mm FPC	2
—	Adapter board	approx. 15 × 12 mm, two-layer	5 (MOQ)
—	Enamelled copper wire	0.1 mm	1 m

Dropped compared to variant A: Q1, R2, R3.

On the PCA9633 package: the TSSOP-16 measures 5.0 × 4.4 mm and hand-solders well. The HVQFN-16 at 3 × 3 mm saves space but has a thermal pad and needs hot air. It is worth checking whether the **PCA9632** is available — same functionality in an 8-pin package, noticeably smaller. Read the datasheet: channel current and address range need to match.

A5. Firmware

Replaces chapter 6. No LEDC, no `gpio_hold` — the driver holds its own state.

```
cpp

#include <Wire.h>

#define PCA_ADDR    0x62
#define FL_MAX_DUTY 204    // 80 %, thermal headroom

static void pcaWrite(uint8_t reg, uint8_t val) {
    Wire.beginTransmission(PCA_ADDR);
    Wire.write(reg);
    Wire.write(val);
    Wire.endTransmission();
}

void flInit() {
    Wire.begin(SDA, SCL);
    pcaWrite(0x00, 0x00);    // MODE1: clear SLEEP bit
    pcaWrite(0x08, 0x2A);    // LEDOUT: LED0..2 to PWM, LED3 off
    flSet(0);
}

void flSet(uint8_t duty) {
    if (duty > FL_MAX_DUTY) duty = FL_MAX_DUTY;
    pcaWrite(0x02, duty);    // PWM0
    pcaWrite(0x03, duty);    // PWM1
    pcaWrite(0x04, duty);    // PWM2
}

void flOffBeforeSleep() {
    flSet(0);
    pcaWrite(0x00, 0x10);    // MODE1: set SLEEP bit, ~1 µA
}
```

A5.1 The deep sleep trap in this variant

The 3.3 V rail stays up during deep sleep, and so does the PCA9633. **It holds its PWM value indefinitely.** Forget `flOffBeforeSleep()` and the frontlight keeps burning while the ESP32 sleeps — the battery is flat in eight hours.

This is the same risk as a missing gate pulldown in variant A, except without a hardware safeguard. `flOffBeforeSleep()` must run before **every** `esp_deep_sleep_start()`.

A6. Additional open items for the clone

The layout analysed here is the official SQFMI design. SQFMI never published layout files for the v2, so the USB-C clone is a redraw. To resolve before soldering:

Item	How
Whether SCL/SDA/3V3/GND really sit on the four pads	measurement per A2.1
Whether the clone follows the v2 pinmap	flash the Watchy library with <code>ARDUINO_WATCHY_V20</code> — if display, all four buttons, vibration, BMA423 and RTC work, the mapping matches
Which ESP32 is fitted	<code>esptool.py chip_id</code> or <code>ESP.getChipModel()</code>
GDEH0154D67 or GDEY	read the marking on the display FPC

The last item affects chapter 3 of the main document: the dimensions given there apply to the GDEY. If the clone carries a GDEH, the case modification has to be re-measured against the actual display. Electrically the two are identical (SSD1681, 24-pin, `GxEPD2_154_D67`).